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Installation Survey

Chassis Model	Sales designation	Body Model	Version	Manufacturer
435.115	U 1300 L	435.500	Box-type body	Waggonfabrik Ürdingen

Technical Data**Body Inside Dimensions**

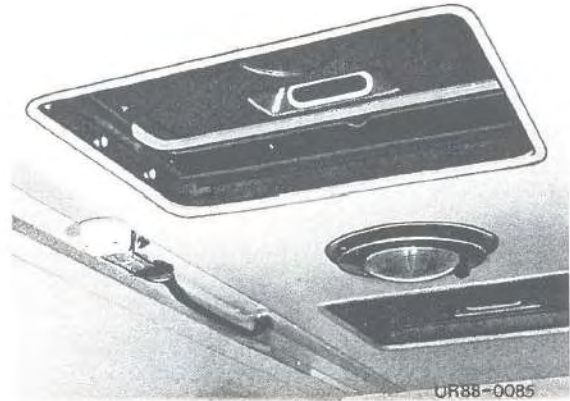
Length inside approx.	mm	3150
Width inside approx.	mm	2170
Height interior approx.	mm	1620

Tightening Torques

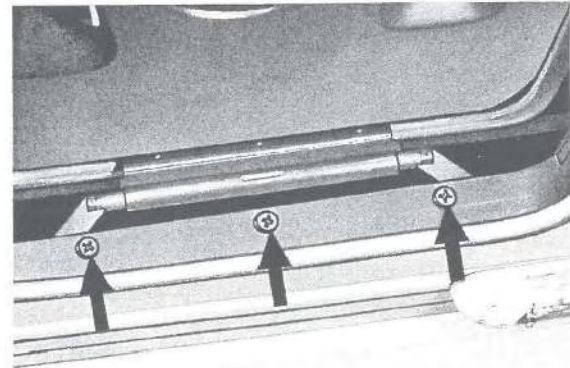
Designation	Thread	Nm
Mount to platform substructure	M 16 x 1,5	160
Bearing shell to frame	M 12 x 1,5	110
Platform bearing bracket to frame front	M 14 x 1,5	195
Mount, front to frame	M 12 x 1,5	115
Mount, rear to frame	M 12 x 1,5	115
Support to platform	M 10	64
Strut to support	M 10	64
Strut to bracket	M 14 x 1,5	195

Exchanging

1 Open roof hatch.



2 Unscrew fastening bolts on both sides.

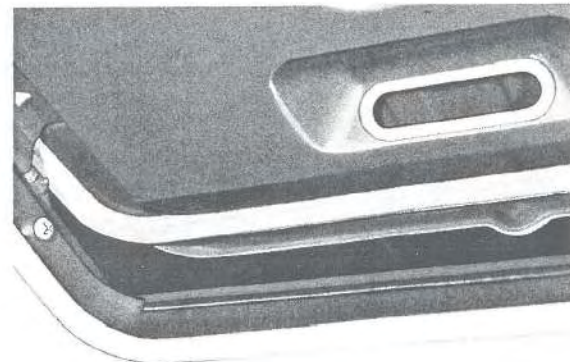


3 Remove roof hatch by pulling obliquely through the opening to the inside.



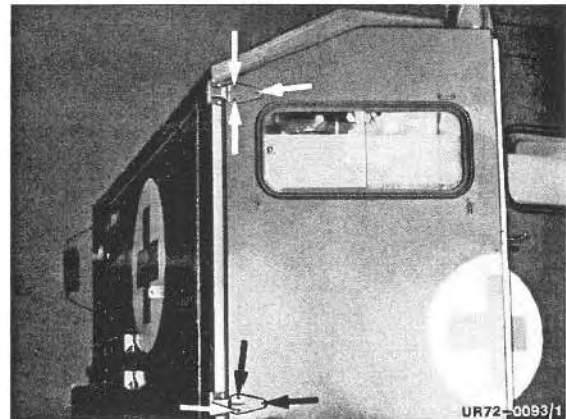
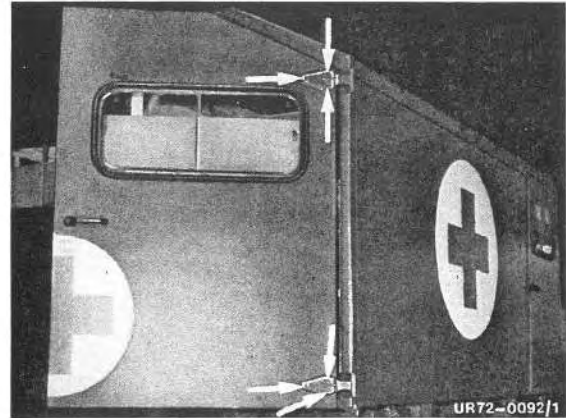
4 Install in reverse order of removal.

Installation note: Align roof hatch so that the rubber seal rests on the edge of the hatch when closed. If necessary drill screw holes.



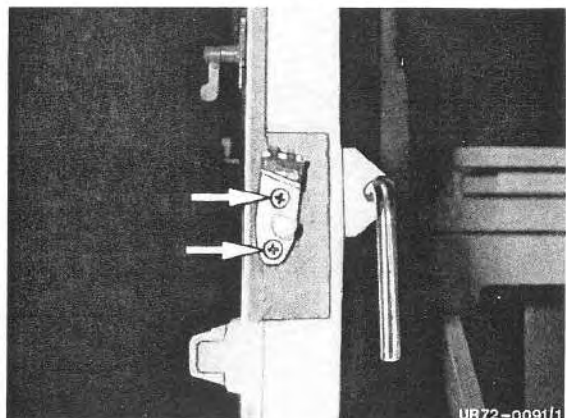
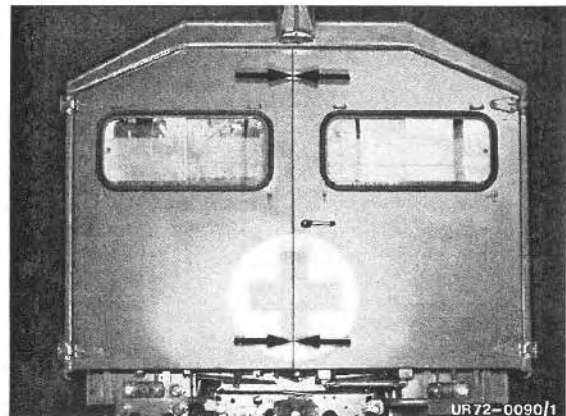
Renewing

- 1 Open right door and unscrew at the hinges.
- 2 Open left door and unscrew at the hinges.
- 3 Install in reverse order of removal.



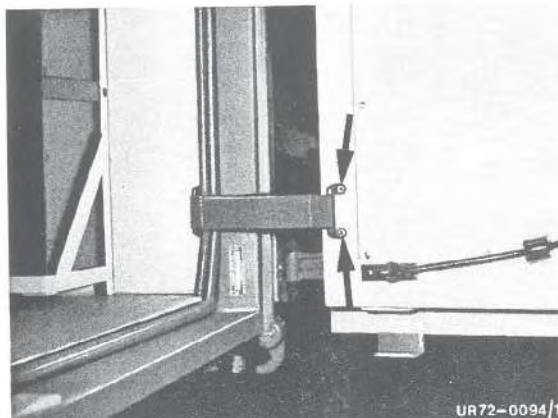
Adjusting

- 1 Close doors and check transition at outer edge. The transition should be uniform and should not exceed 1 mm.
- 2 If door setting needs to be changed, release striker plate and vary height or to the side correspondingly.
- 3 Tighten striker plate and check function of door lock.

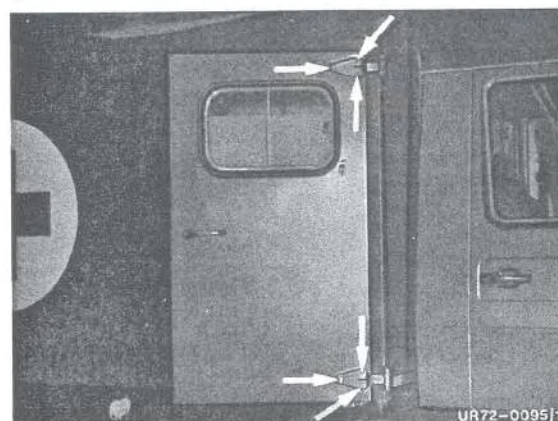


Renewing

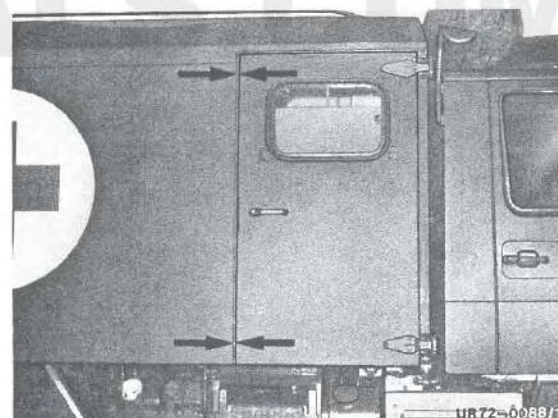
- 1 Open side door and unscrew catch strip.



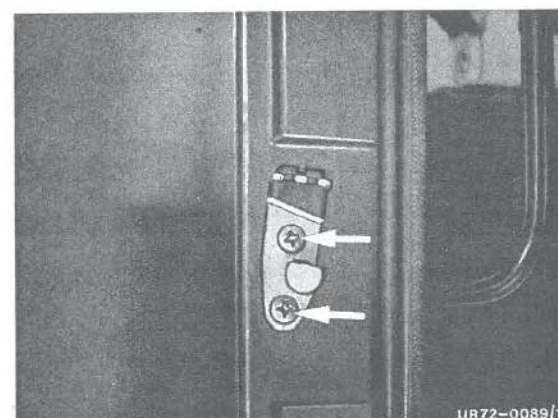
- 2 Unscrew side door at hinges.
- 3 Install in reverse order of removal.

**Adjusting**

- 1 Close doors and check transition at outer edge. The transition should be uniform and must not exceed 1 mm.

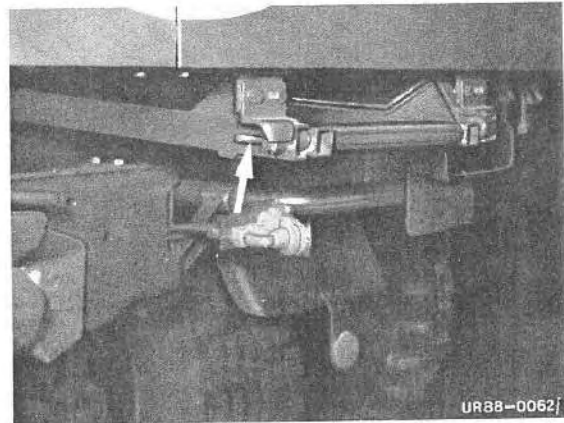


- 2 If the door setting needs to be changed, release striker plate and vary height or to the side correspondingly.
- 3 Tighten striker plate and check function of door lock

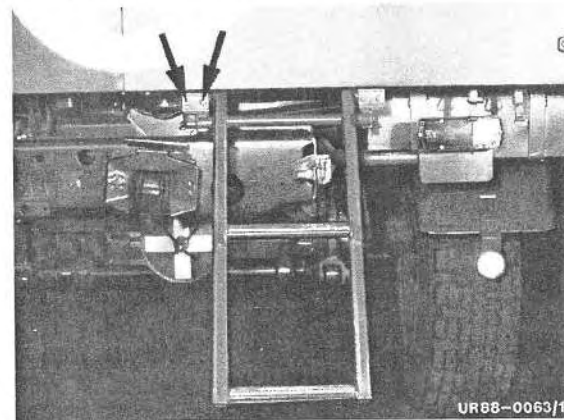


Renewing

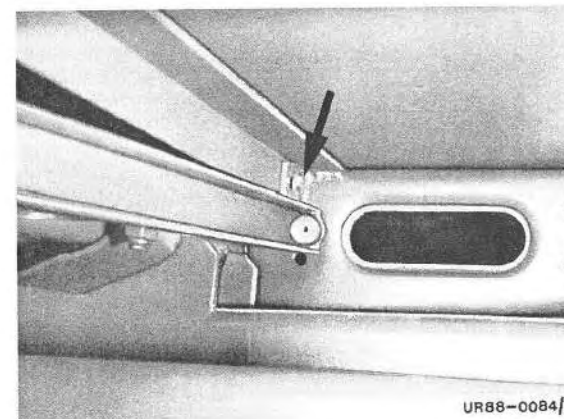
1 Open locking pin and withdraw steps.



2 Unscrew guide rail at rear.

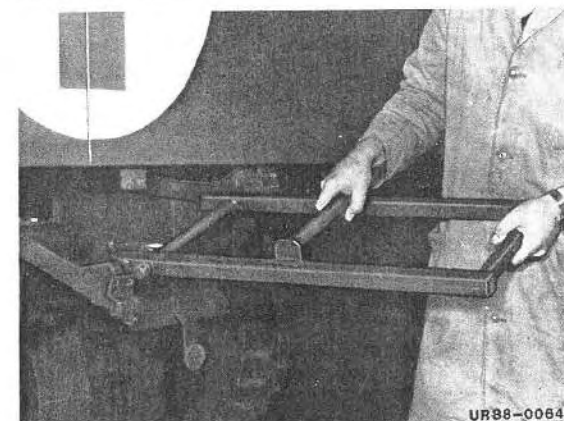


3 Unscrew guide rail at front.



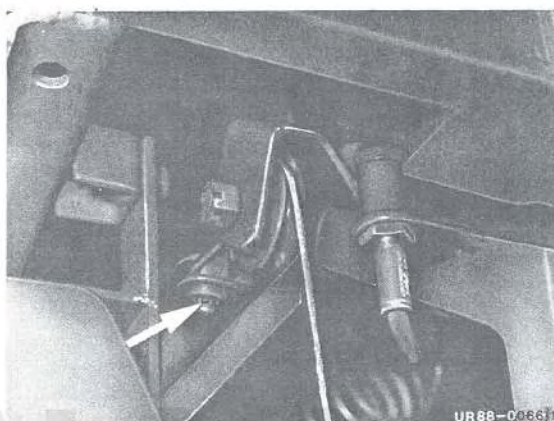
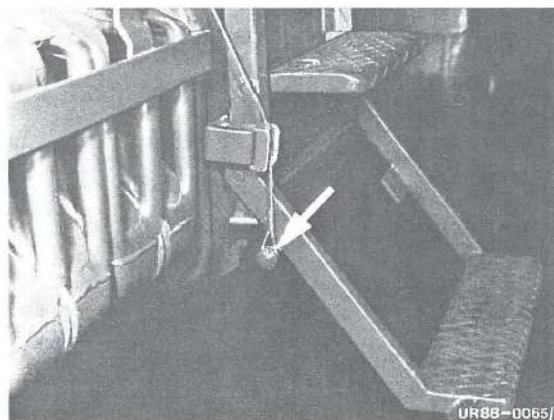
4 Detach steps from guide rail and remove.

5 Install in reverse order of removal.



Renewing

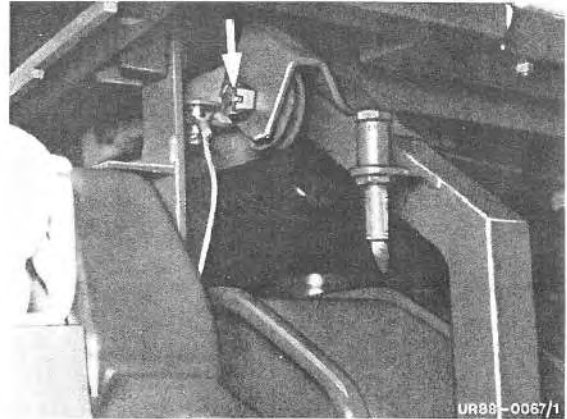
- 1 Open side door and withdraw steps.
- 2 Detach cable at bottom.
- 3 Release cylinder pin at top and remove.
- 4 Remove cable.
- 5 Install in reverse order of removal.
- 6 Check that the roller pulley is free moving, release if necessary.



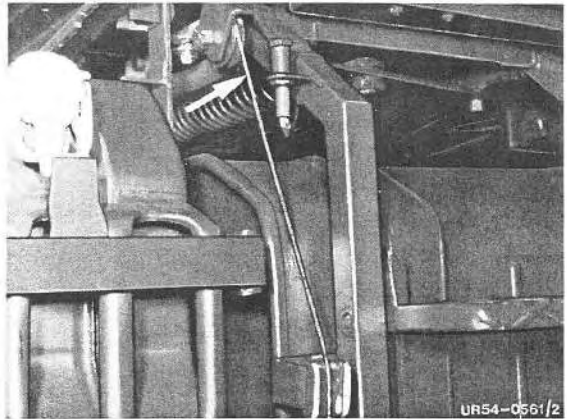
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Renewing

- 1 Open side door and withdraw steps.
- 2 Remove cable from roller pulley.
- 3 Release roller pulley, unscrew and remove.
- 4 Exchange wearing parts.
- 5 Install in reverse order of removal.



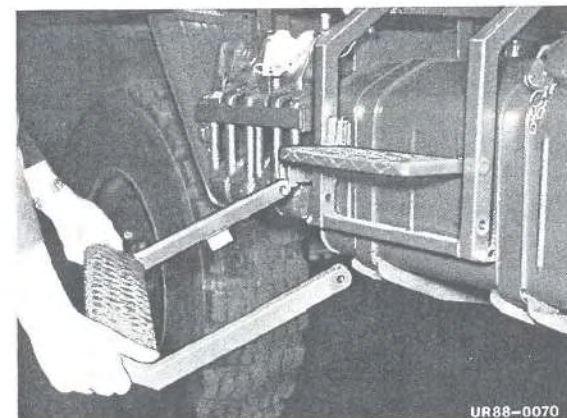
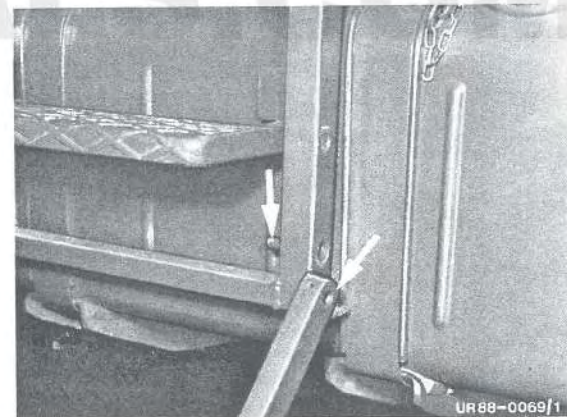
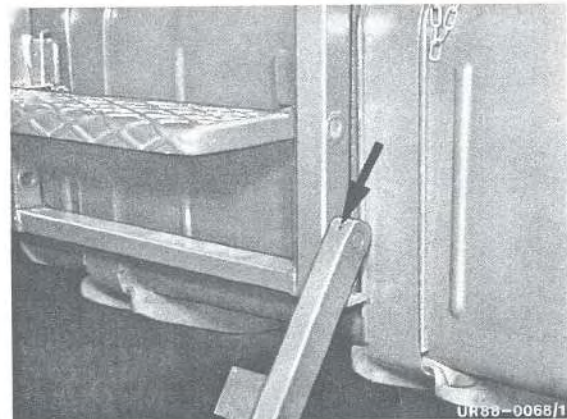
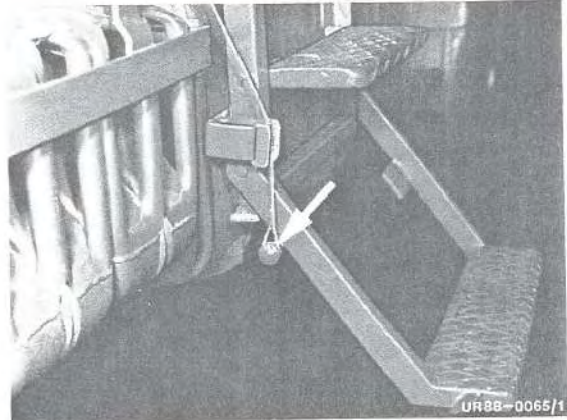
UR88-0067/1



UR54-0561/2

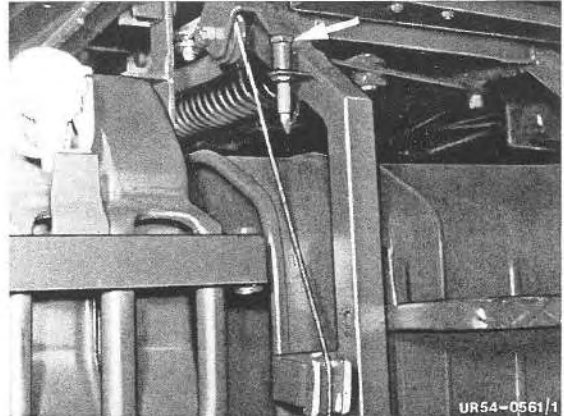
Renewing

- 1 Detach cable.
- 2 Drive out dowel pin.
- 3 Drive out pivot pin.
- 4 Remove folding step.
- 5 Install in reverse order of removal.

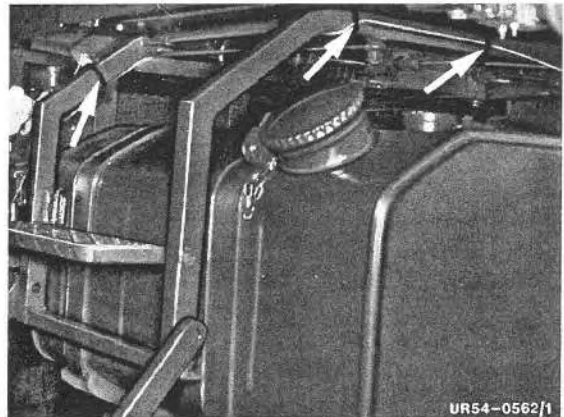


Removal and Installation

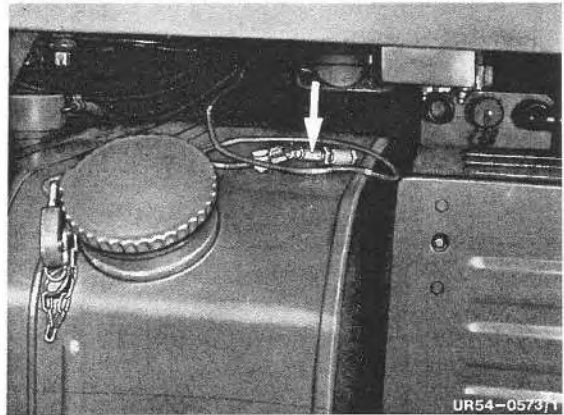
- 1 Relieve supply pressure.
- 2 Detach induction sensor.



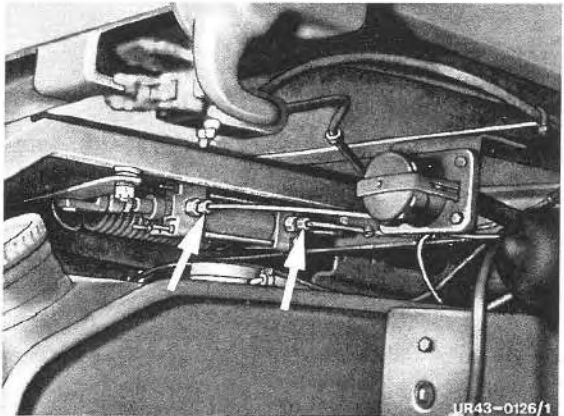
- 3 Cut off cable clips.



- 4 Remove induction sensor with cable and place to one side.

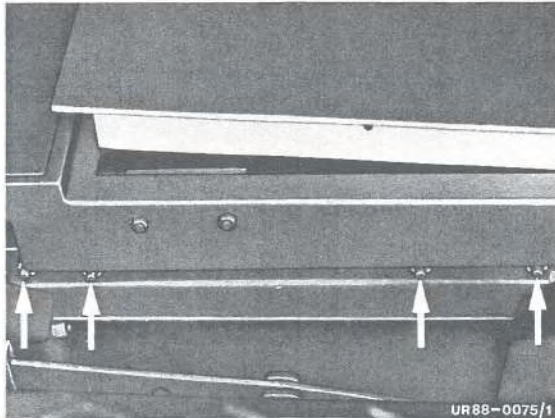


- 5 Detach and mark compressed air hoses on working cylinder.

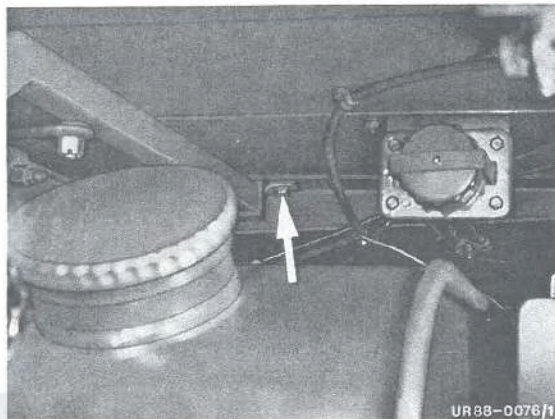


60.12 Removal and Installation of Side Entrance Steps

435.500



6 Unscrew frame from floor at front.



7 Unscrew frame bracket from floor at rear.



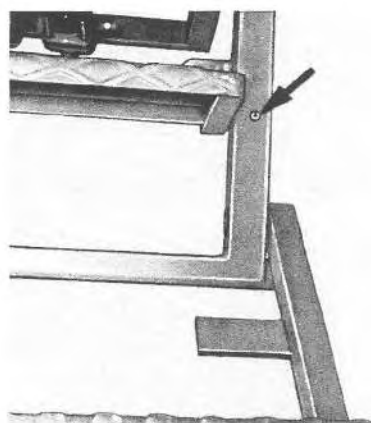
8 Completely remove steps from vehicle.

9 Install in reverse order of removal.

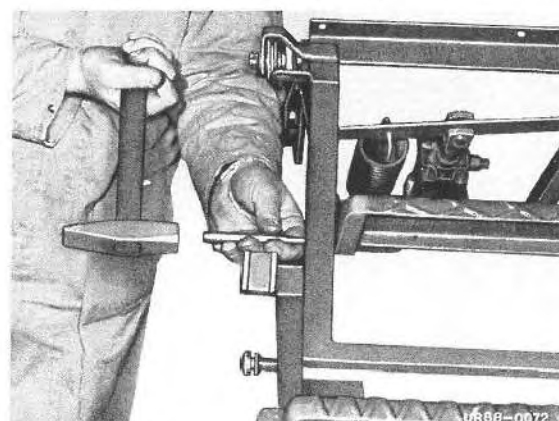
10 Adjust induction sensors, see 82.12– 7.1/1.

Renewing

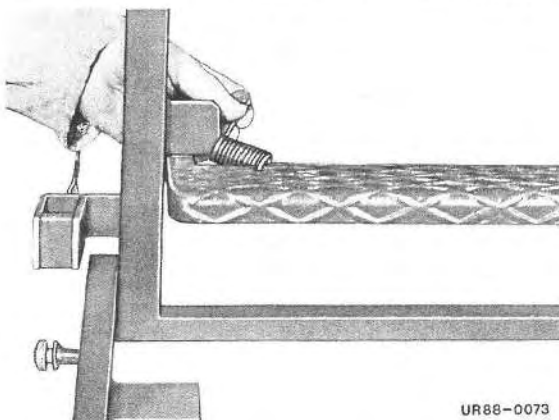
- 1 Completely remove side entrance steps, see 9.1/1.
- 2 Drive out dowel pin.
- 3 Drive shaft up to the center.
- 4 Remove spring.
- 5 Completely drive out shaft and remove folding step.
- 6 Install in reverse order of removal.



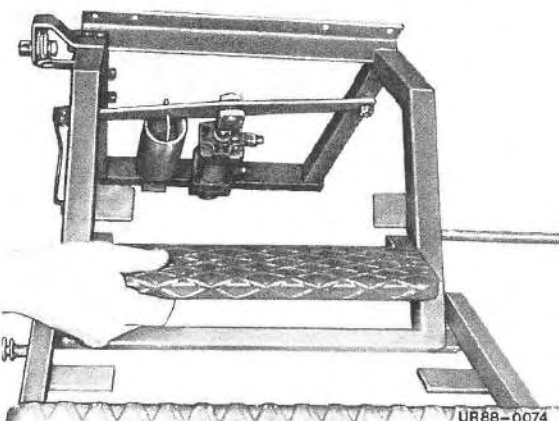
UR88-0071/1



UR88-0072



UR88-0073

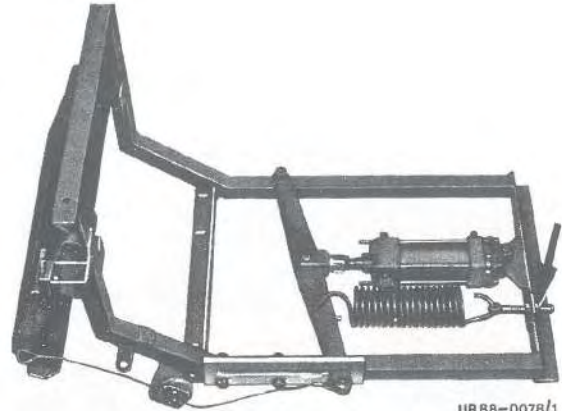


UR88-0074

Removal and Installation

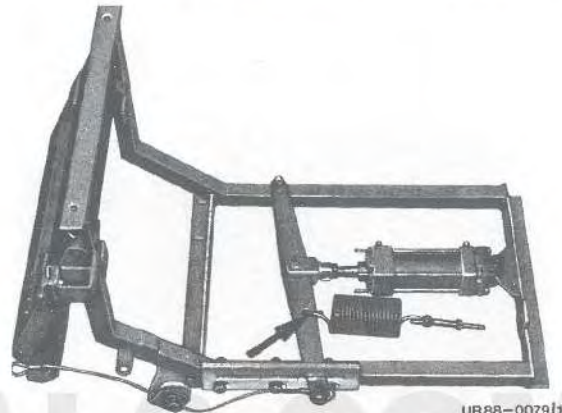
1 Completely remove side entrance steps, see 9.1/1.

2 Unscrew tensioning bolt.



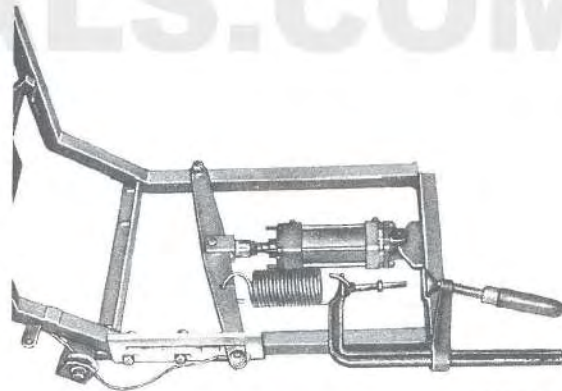
UR88-0078/1

3 Detach return spring from return arm.



UR88-0079/1

4 Install in reverse order of removal.

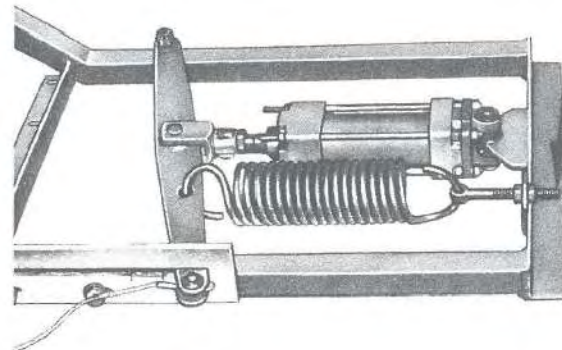


UR88-0080

Installation note

Clamp return spring with screw clamp until tensioning bolt can be secured in position.

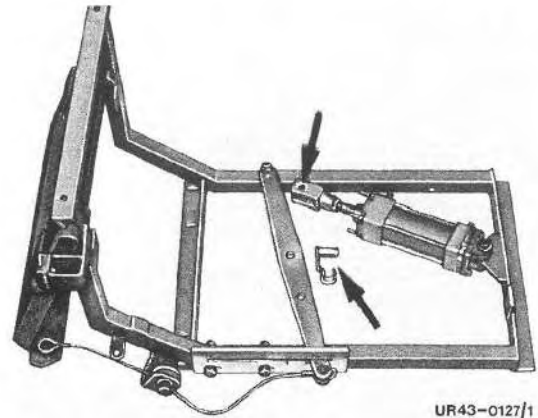
– The threads of the tensioning bolt should end with the frame profile.



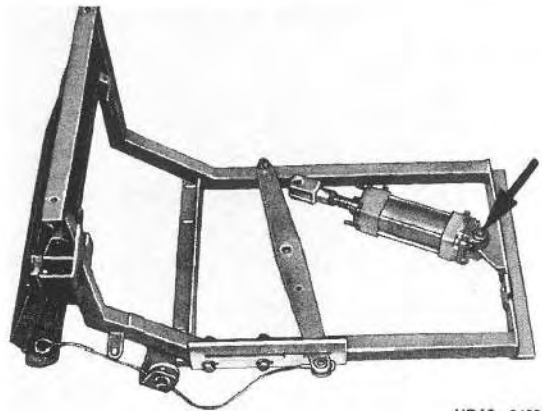
UR88-0081/1

Removal and Installation

- 1 Completely remove side steps, see 9.1/1.
- 2 Remove return spring, see 11.1/1.
- 3 Release and remove piston rod.
- 4 Release and remove bottom cylinder pin.
- 5 Remove cylinder.
- 6 Install in reverse order of removal.



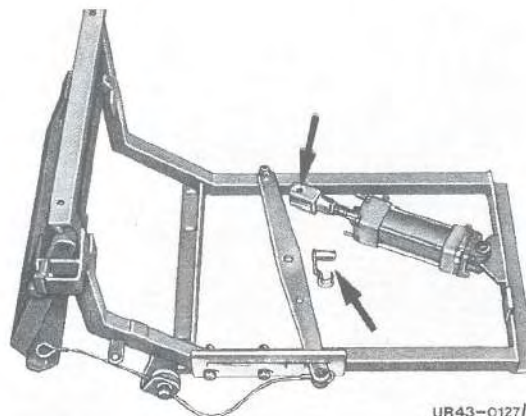
UR43-O127/1



UR43-O128/1

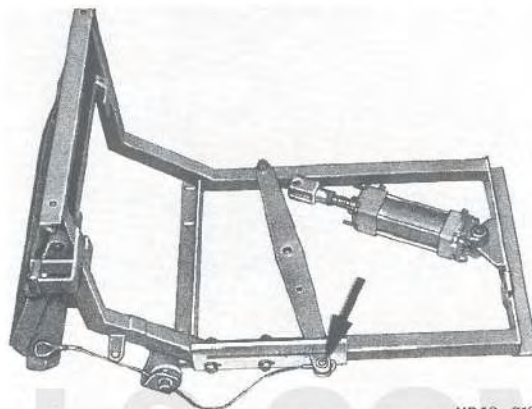
Removal and Installation

- 1 Completely remove side steps, see 9.1/1
- 2 Remove return spring, see 11.1/1.
- 3 Release and remove piston rod.



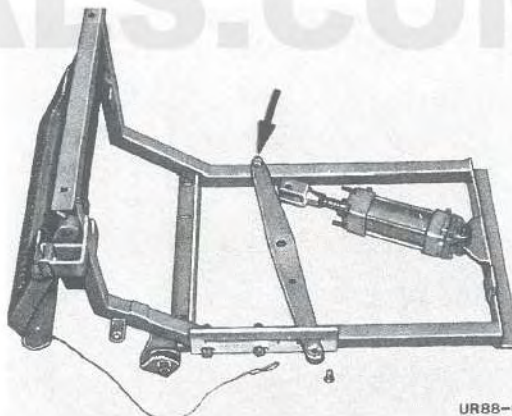
UR43-0127/1

- 4 Release and remove cable.



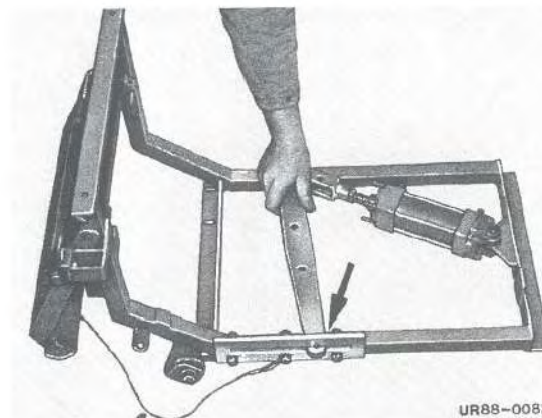
UR43-0128/2

- 5 Release and unscrew return arm.



UR88-0082/1

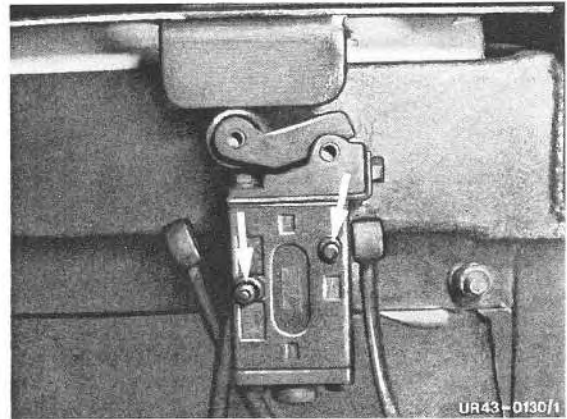
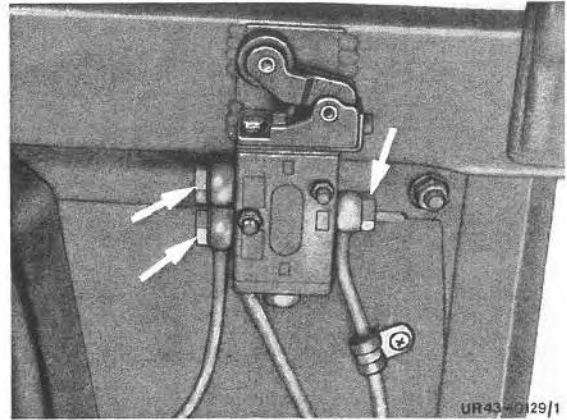
- 6 Remove return arm through cut-out in plastic plate.
- 7 Install in reverse order of removal.



UR88-0083/1

Renewing

- 1 Relieve supply pressure at compressed air tank.
- 2 Unscrew compressed air lines.
- 3 Unscrew and remove 3/2-way valve.
- 4 Install in reverse order of removal.
- 5 Check function and for leaks.



Renewing

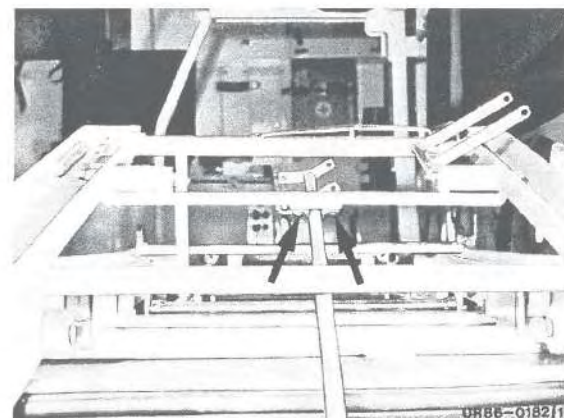
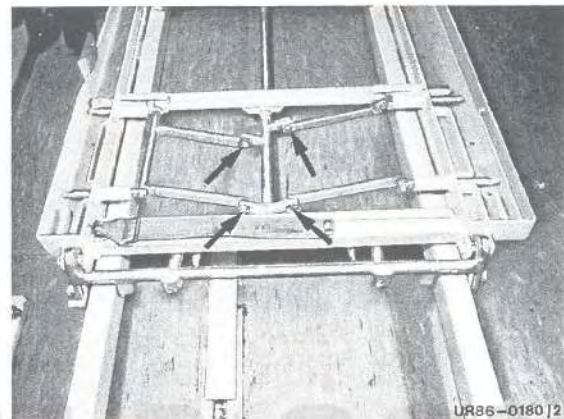
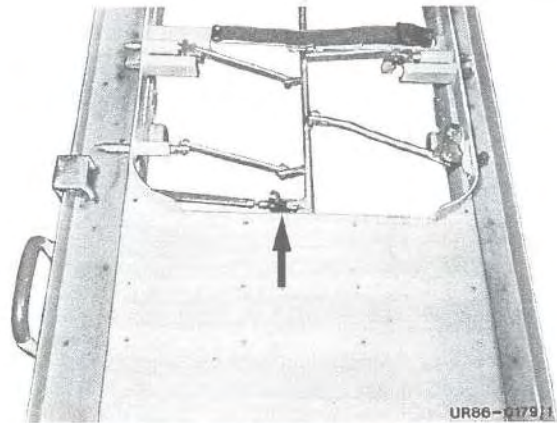
1 Withdraw stretcher holder to rear and remove stretcher.

2 Release locking shaft at rear.

3 Remove retainers on locking shaft at front.

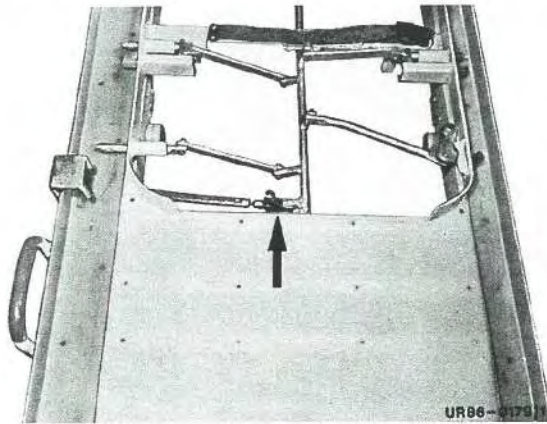
4 Unscrew guide bearing at rear.

5 Unscrew guide bearing at front and remove locking shaft.



60.12 Renewing Locking Shaft for Stretcher Holder

435.500

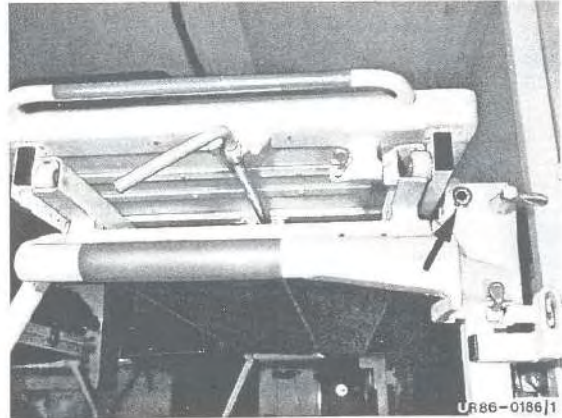


6 After installing the locking shaft, adjust clevis by turning so that the locking lever can be heard to engage closing.

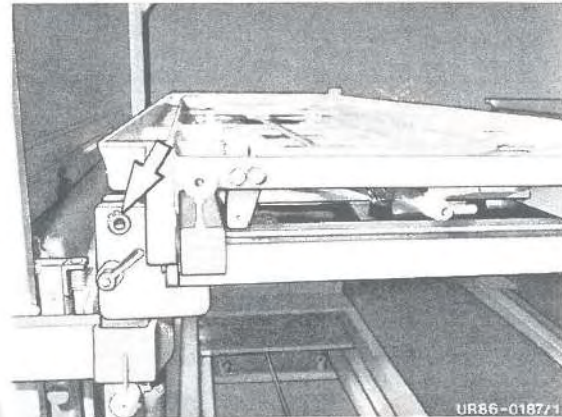
7 Secure clevis.

Removal and Installation

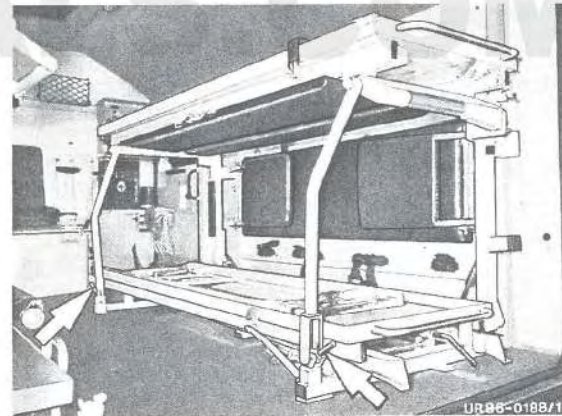
1 Release pin at rear of supporting arm joint and remove.



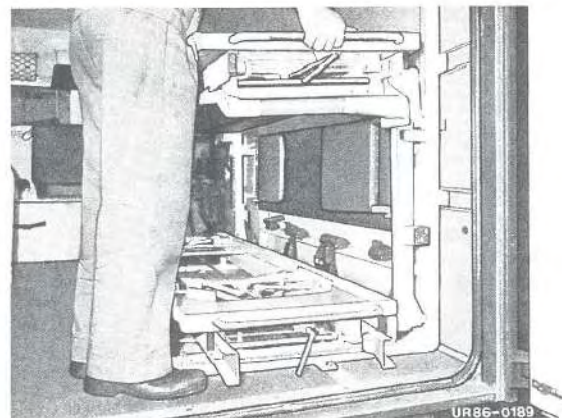
2 Release pin at front of supporting arm joint and remove.



3 Unlatch both vertical supports and swivel upwards.

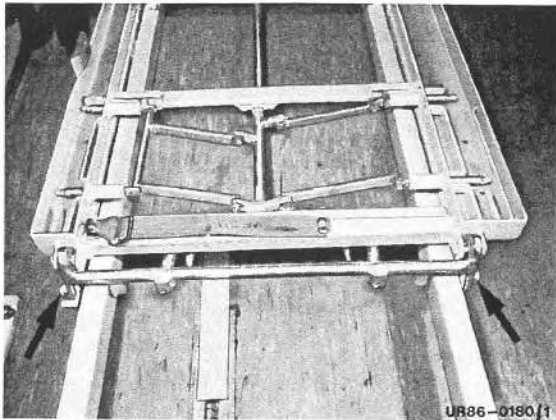


4 Remove top stretcher holder.



60.12 Removal and Installation of Stretcher Frame

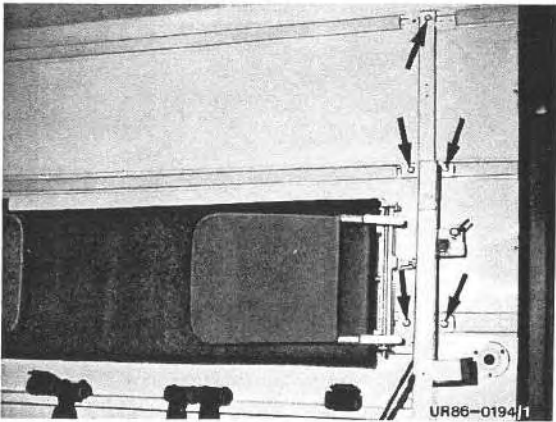
435.500



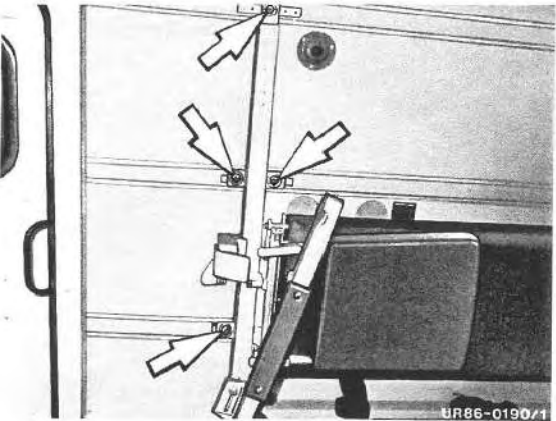
5 Unscrew limit stops from lower stretcher holder and detach springs.

Note: Pay attention to shims.

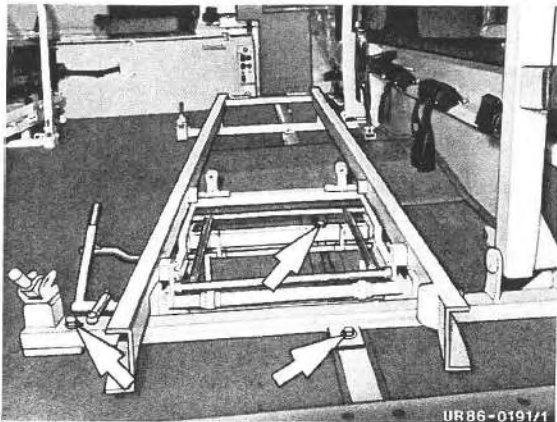
6 Pull stretcher holder to rear and remove.



7 Unscrew stretcher brace at rear of side wall.

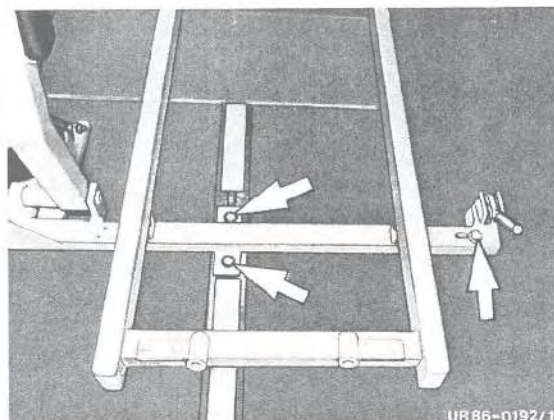


8 Unscrew stretcher brace at front.



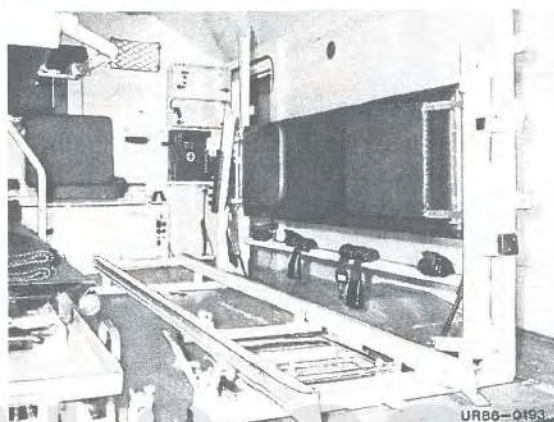
9 Release bottom fastening bolts.

- 10 Unscrew stretcher brace at front of floor.



- 11 Remove stretcher frame out of box-type body.

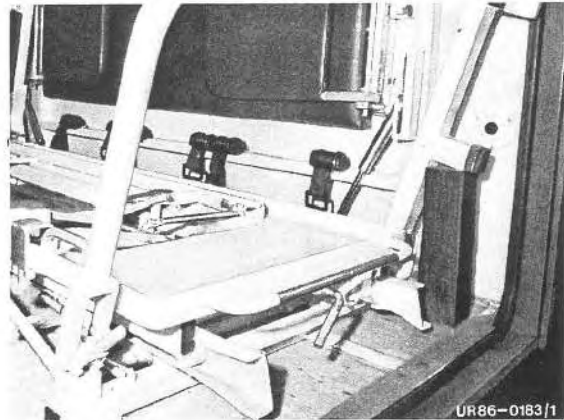
- 12 Install in reverse order of removal.



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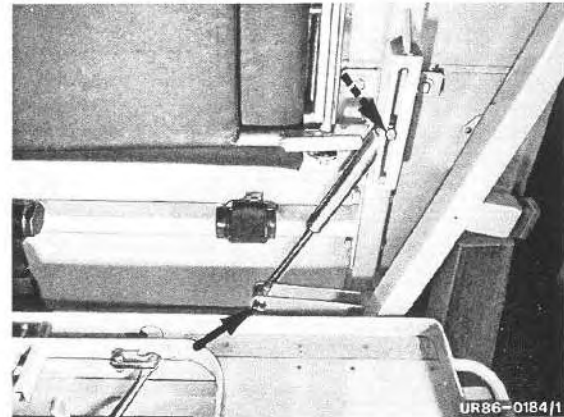
Removal and Installation

1 Pull top stretcher holder to the rear (approx. 30 cm) and secure against tilting.



2 Release and remove rear pressurized spring.

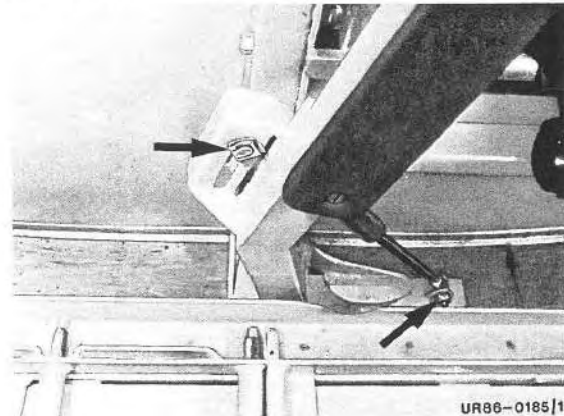
Note: Pay attention to spacer bushings.



3 Release and remove front pressurized spring.

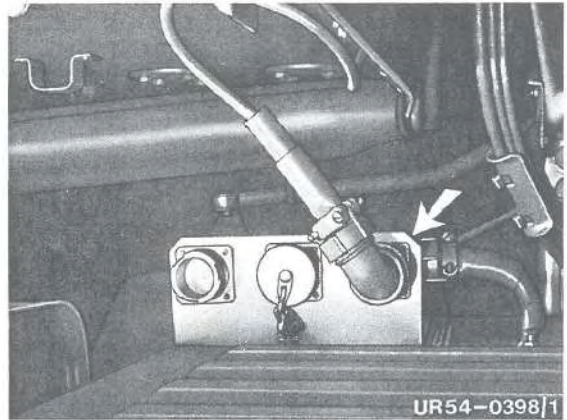
Note: Pay attention to top spacer bushings.

4 Install in reverse order of removal.

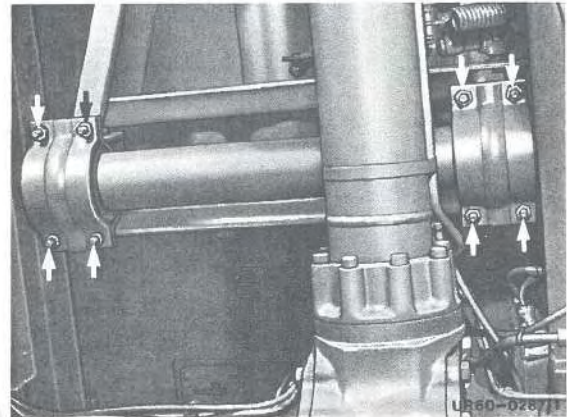


Removal and Installation

1 Release central cable connection between box body and chassis and remove.



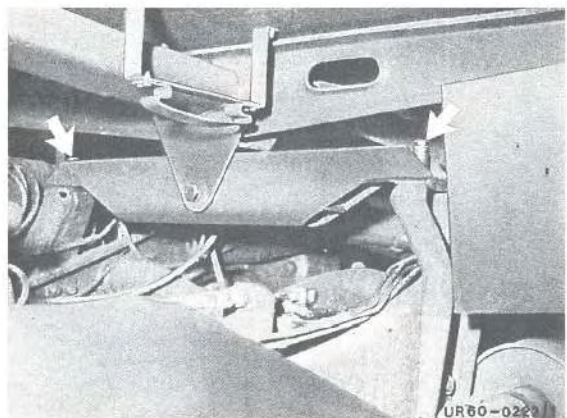
2 Unscrew both bearing caps to the left and right of the centre platform support and remove.



3 Unscrew rear cross member to the left and right of the bearing block.



4 Unscrew front cross members at frame.



60.12 Removal and Installation of Box-Type Body

435.500

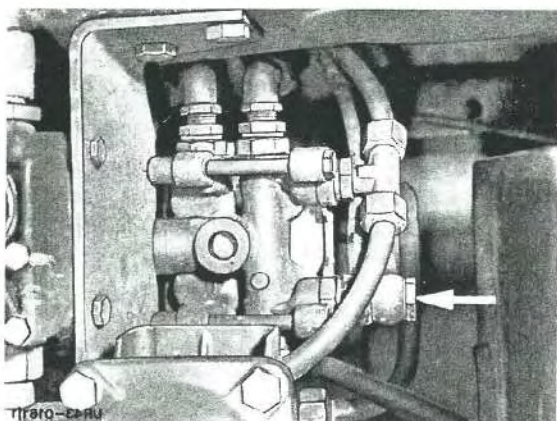


5 Unscrew front section of the left rear fender.

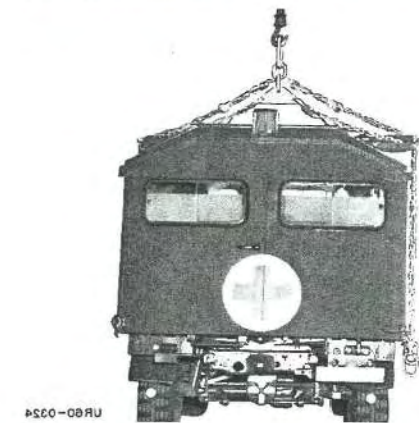
6 Unscrew fuel line for auxiliary box body heater at the sending unit, catch fuel.

7 Open tank cover.

Note: If the tank cap is not opened, the tank may drain.



8 Unscrew delivery line at connection 24 of the four-circuit protection valve.



9 Place lifting tackle in position and remove box body.

10 Install in reverse order of removal.
Tightening torques, see 1.1/1.



Installation note: When mounting the box body, ensure the rubber bearing shells are correctly seated on the center platform support.

Survey

Version

Electrical System (Driver's Cab)

82.12

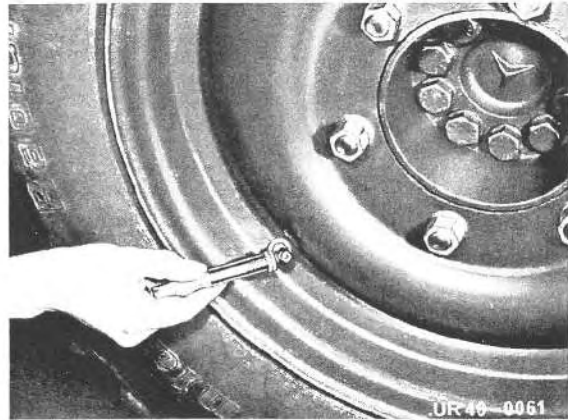
Electrical System (Box-Type Body)

82.15

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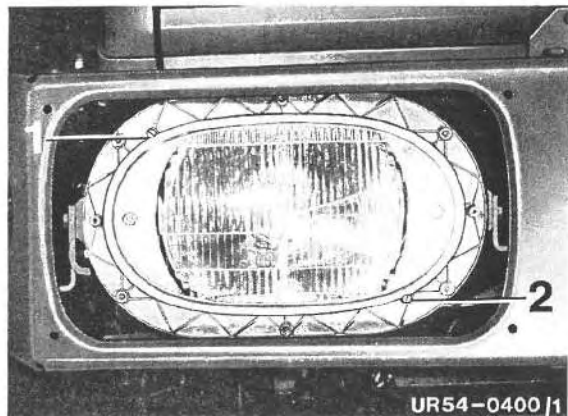
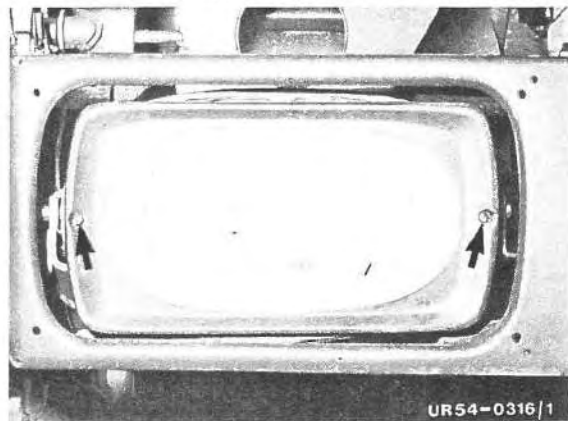
Checking

- 1 Check tire pressure, correcting if necessary.
- 2 Park vehicle on flat surface.
- 3 Prepare beam setter with reference to manufacturer's instructions.
- 4 Check headlight settings.



Adjusting

- 1 Unscrew protective stone grille (only with SA 35 770)
- 2 Unscrew headlamp moulding.
- 3 Adjust headlight at vertical adjusting screw (2) or lateral adjusting screw (1) by corresponding amount.
- 4 Screw on headlight molding.



Technical Data Chassis (435.115) for Box-Type Body**Indicator Light for Folding Step Series 435.500**

Voltage	V	24
Power output	W	4

Rotating beacon - Halogen front

Designation		SA 35 941/08
Voltage	V	24
Output	W	70
Test symbol		~ K 8605
Quantity		2

Alarm Switch for RK Light and Special Signal (Instrument Panel)

Designation		SA 35 941/08
Voltage	V	24
Current	A	10

Sound Sequence - Control Unit

Designation		SA 35 941/08
Voltage	V	24
Type		5 AB 003 188 - 01
Designation		DIN 14 610
Test symbol		~ B 09

Tachograph (With Auxiliary Recorder for Special Purpose Vehicles)

Designation		SA 35 771/08
Manufacturer		Messrs. Kienzle
Type		TCO 15 - 2
Voltage	V	24
Approval PTB		334 500
Test symbol		~ B 122
Drive constant		1000 rev/km

Search lamp

Designation		SA 36 038/01
Voltage	V	24
Power output	W	20

Circuit Diagram (from SP)

Key to 1.3/3

1 Power supply, starting control

- G 1,2 Batteries
- G 3 Alternator
- H 1 Indicator lamp charging indicator
- M 1 Starter
- P 1 Regulator
- Q 1 Battery main switch
- Q 2 Drive switch
- X 1 Assisted start socket

2 Power supply and call system

- F 1 Overcurrent protection switch, power supply
- F 2 Overcurrent protection switch, call system
- H 2 Buffer call system
- H 3 Indicator lamp, call system
- H 4 Indicator lamp, folding step
- K 1 Isolating relay, power supply
- K 2 Relay, power supply
- X 2 Socket, power supply and call system

3 Headlamp, IR - light, brake system

- F 3 Overcurrent protection switch, low beam left
- F 4 Overcurrent protection switch, high beam left
- Indicator lamp, high beam
- F 5 Overcurrent protection switch, low beam
- Right, rear fog light (SA)
- F 6 Overcurrent protection switch, high beam right
- F 7 Overcurrent protection switch, IR - auxiliary light and indicator lamp
- F 8 Overcurrent protection switch, stop lamps
- Masked stop lamps, horn, turn signal and hazard warning system, special signal
- H 5 Headlamp, left
- H 6 Indicator lamp, high beam
- H 7 Headlamp, right
- H 8 Indicator lamp, IR - auxiliary light
- H 9 IR - auxiliary light
- H 10 Stop light, right
- H 11 Stop light, left
- H 12 Masked stop lamp
- S 1 High - low beam switch
- S 2 Switch IR - auxiliary light
- S 3 Sending unit stop light
- S 4 Main masked light switch
- X 3 Socket, 7 - pole (SA)
- X 4 Socket, 12 - pole

4 Headlight flasher, interior lamp, reversing lamp

- F 9 Overcurrent protection switch, shift gate indicator
- Reversing lamp, headlight flasher
- Interior lamp
- H 13 Reversing lamp (SA)
- H 14 Interior lamp
- K 3 Relay masked light
- S 5 Headlight flasher
- S 6 Sending unit, reversing lamp
- S 7,8 Switch, interior light

5 Horn

- H 15 Horn
- S 9 Switch horn
- X 5 Connection alarm switch

6 Turn signal - hazard warning system

- H 16 Indicator lamp, turn signal tractor
- H 17 Turn signal, rear left
- H 18 Turn signal, front left
- H 19 Turn signal, front right
- H 20 Turn signal, rear right
- H 21 Indicator lamp, turn signal trailer
- H 22 Indicator lamp, hazard warning system
- P 2 Sending unit, hazard warning system
- S 10 Hazard warning flasher switch
- S 11 Turn signal switch
- X 3 Socket, 7 - pole (SA)
- X 4 Socket, 12 - pole

7 Wiper and washer unit

- F 10 Overcurrent protection switch, reserve
- F 11 Overcurrent protection switch, windscreen washer
- F 12 Overcurrent protection switch, windscreen wiper
- M 2 Windscreen washer pump
- M 3 Wiper motor
- S 12 Switch, windscreen washer
- S 13 Switch, windscreen wiper
- A = A = ON/OFF, B = slow - fast

8 Heating

- F 13 Overcurrent protection switch heater fan
- M 4 Heater fan
- S 14 Switch heater fan

9 Operation monitoring

- A 1 Diode group
- B 1 Sending unit, coolant temperature indicator
- B 2 Sending unit, oil pressure gauge
- B 3 Sending unit, fuel gauge
- F 14 Overcurrent protection switch, instruments
- Indicator lamps
- H 23 Indicator lamp, differential pressure gauge
- H 24 Indicator lamp, brake pad wear indicator RA
- H 25 Indicator lamp, brake pad wear indicator FA
- H 26 Indicator lamp, parking brake
- H 27 Indicator lamp, four - wheel drive
- H 28 Indicator lamp, differential lock FA/RA
- H 29 Indicator lamp, supply pressure
- H 30 Revolution counter
- H 31 Instrument cluster
- S 15 Sending unit, differential pressure gauge
- S 16 Sending unit, brake pad wear indicator RA
- S 17 Sending unit, brake pad wear indicator FA
- S 18 Sending unit, parking brake
- S 19 Sending unit, four - wheel drive
- S 20 Sending unit, differential lock

54.13 General

SA 35 941/17

Key to 1.3/3 (cont'd.)

- V 1 Diode, differential pressure gauge
- V 2 Diode, brake pad wear indicator RA
- V 3 Diode, brake pad wear indicator FA

10 Side - marker lamp, license plate light

Tail light, instrument illumination

- F 15 Overcurrent protection switch, side - marker lamp
Left indicator lamp, rear fog light
- F 16 Overcurrent protection switch side - marker lamp
Right, relay shift indicator
- F 17 Overcurrent protection switch license plate light
Tail light left
- F 18 Overcurrent protection switch, instrument
illumination, tail light right
- H 32 Side - marker lamp left
- H 33 Rear fog light (SA)
- H 34 Side - marker lamp right
- H 35 - 38 Shift gate indicator
- H 39, 40 License plate light
- H 41 Tail light left
- H 42 Tail light right
- K 4 Relay shift indicator (bright - dark)
- S 21 Switch rear fog light (SA)
- S 22 Sending unit, shift gate indicator
- S 23 Regulating switch instrument illumination
- X 3 Socket, 7 - pole (SA)
- X 4 Socket, 12 - pole

11 Special signal, rotating beacon

- A 2 Tone sequence control unit, special signal
- H 43 Rotating beacon, right
- H 44 Indicator lamp, rotating beacon right
- H 45 Indicator lamp, rotating beacon left
- H 46 Rotating beacon, left
- H 47 Special signal, deep tone
- H 48 Special signal, super tone
- K 5 Control relay rotating beacon
rear
- S 24 Alarm switch, special signal with
indicator lamp rotating beacon
rear

12 Speedometer, map light, sockets

- F 19 Overcurrent protection switch speedometer
Map light
- F 20 Overcurrent protection switch socket
Driver's cab
- F 21 Overcurrent protection switch socket
2 - pole
- H 49 Speedometer
- H 50 Map light
- S 25 Switch map light
- X 4 Socket, 12 - pole
- X 6 Socket, driver's cab
- X 7 Socket, 2 - pole

13 Masked light

- F 22 Overcurrent protection switch, convoy cross light
- F 23 Overcurrent protection switch, masked headlight
left
- F 24 Overcurrent protection switch, masked headlight
right
- F 25 Overcurrent protection switch, masked tail light
- H 51 Convoy cross light
- H 52 Masked light headlamp left
- H 53 Masked light headlamp right
- H 54 Masked light tail light right
- H 55 Masked light tail light left
- X 4 Socket, 12 - pole

14 Sockets

- X 3 Socket 7 - pole (SA)
- X 4 Socket 12 - pole

- AI-GI = Plug - and - socket connection driver's cab
- VD = Plug - and - socket connection distributor box,
rear
- X = Ground bulkhead
- Y = Ground frame
- Z = Ground driver's cab